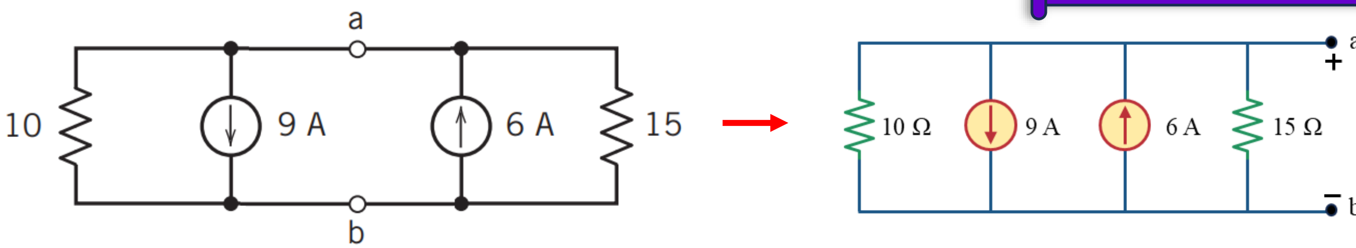
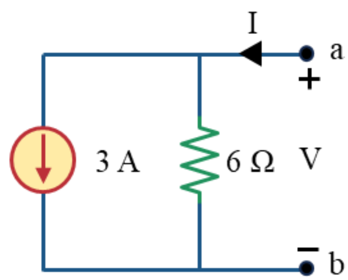
Solution to Problem 1

Did nothing but to bring point a and b at the rightside for better visualization



Calculating the equivalent current source and equivalent resistance in the circuit, we get the circuit below. We also defined V and I in the circuit.



Applying KCL at node a,

$$\boxed{I = 3 + \frac{V}{6}}$$

To draw the graph, let's find two points (because it is a linear circuit and if we can find 2 points, we can draw a straight line through them to draw the IV characteristics)

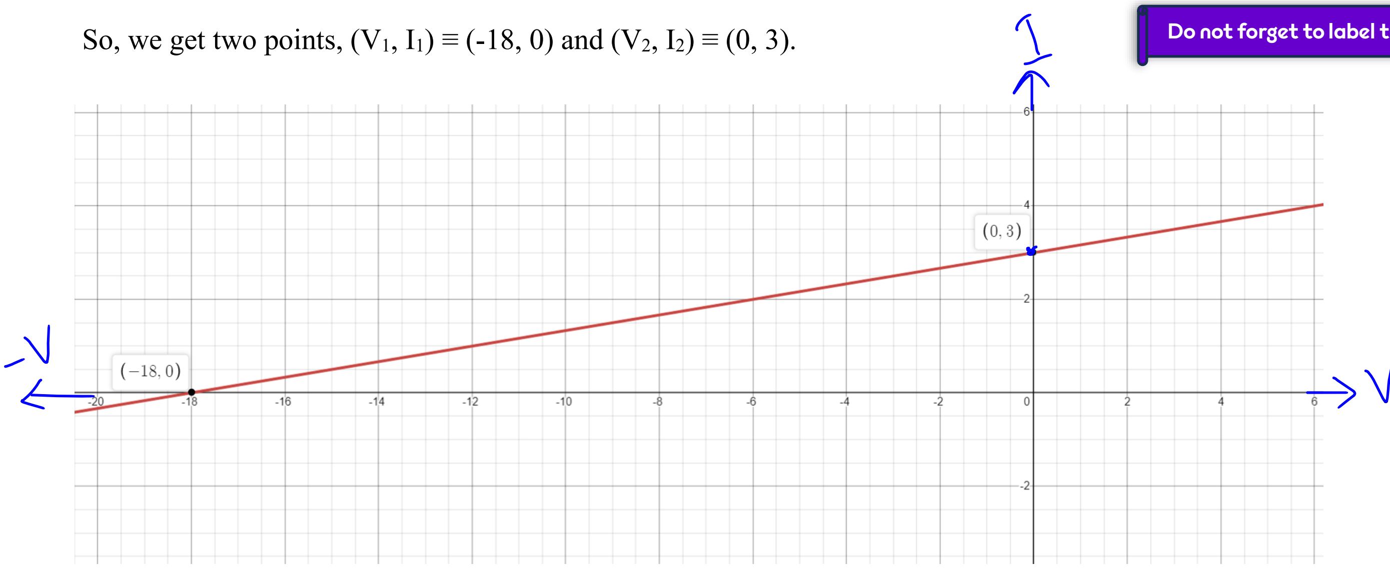
From the equation,

$$if\ I = 0, V = -18$$

$$If\ V = 0, I = 3$$

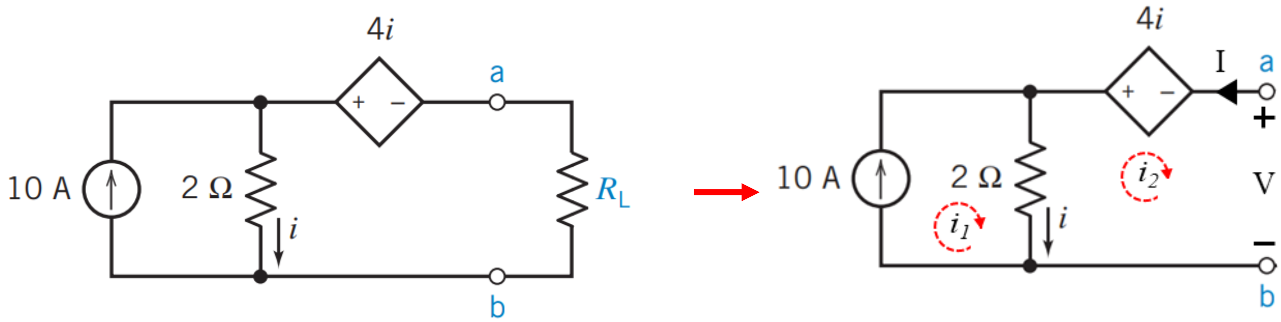
So, we get two points, $(V_1, I_1) \equiv (-18, 0)$ and $(V_2, I_2) \equiv (0, 3)$.

Do not forget to label the axis



We have removed the R_L resistor because we needed only the left portion of the circuit to find the IV characteristics.

Solution to Problem 2



We can see that there are 2 meshes. From the circuit,

$$i_2 = -I$$

$$i = i_1 - i_2$$

Applying KVL in mesh 1,

$$i_1 = 10\ A$$

Applying KVL in mesh 2,

$$2(i_2 - i_1) + 4i + V = 0$$

$$\gg 2(i_2 - i_1) + 4(i_1 - i_2) + V = 0$$

$$\gg 2i_2 - 2i_1 + 4i_1 - 4i_2 + V = 0$$

$$\gg 2i_1 - 2i_2 + V = 0$$

$$\gg 2 \times 10 - 2(-I) + V = 0$$

$$\gg 20 + 2I + V = 0$$

$$\boxed{\gg I = -\frac{V}{2} - 10}$$

To draw the graph, let's find two points (because it is a linear circuit and if we can find 2 points, we can draw a straight line through them to draw the IV characteristics)

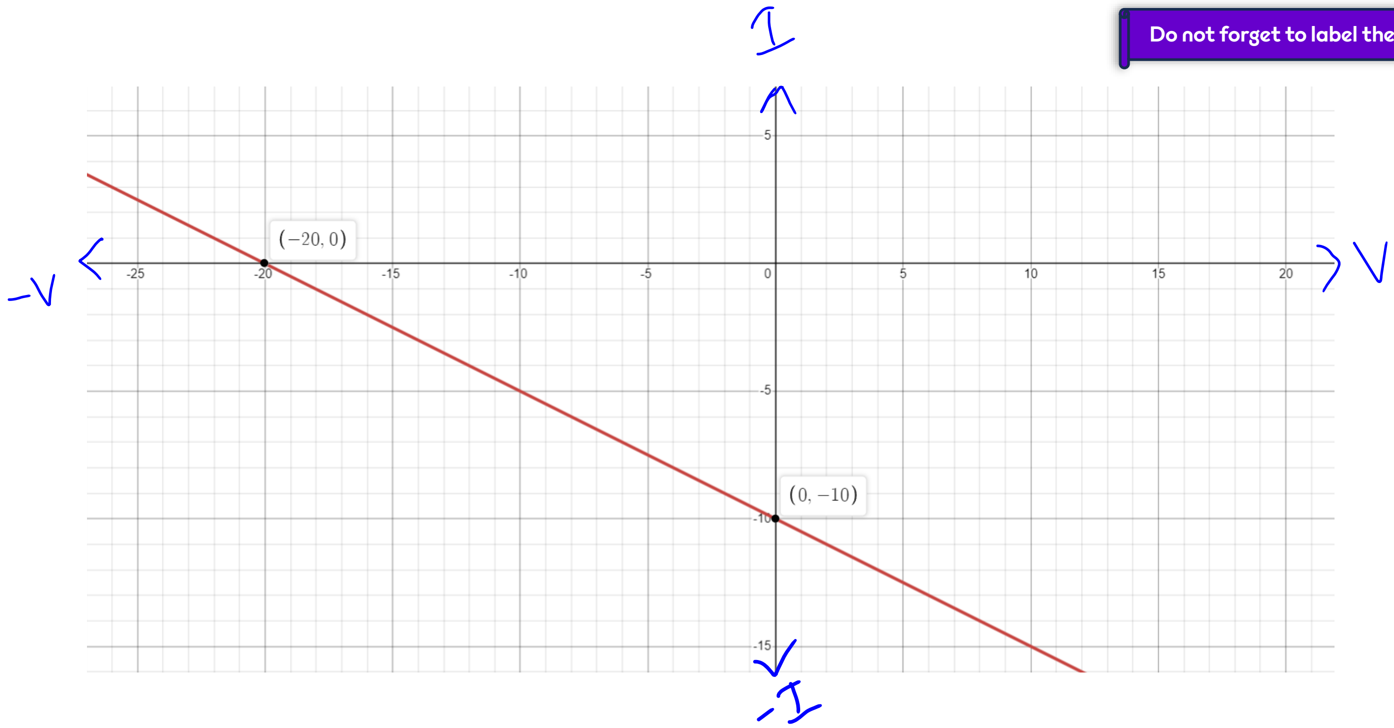
From the equation,

$$if\ I = 0, V = -20$$

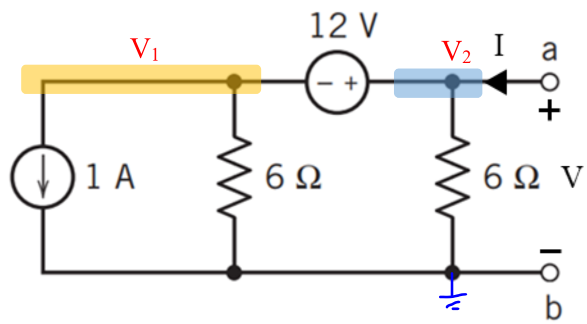
$$If\ V = 0, I = -10$$

So, we get two points, $(V_1, I_1) \equiv (-20, 0)$ and $(V_2, I_2) \equiv (0, -10)$.

Do not forget to label the axis



Solution to Problem 3



To find the IV of this circuit, let us implement nodal analysis. From the circuit,

$$V_2 = V$$

We can see that there is a supernode between node 1 and 2. Applying KCL at supernode,

$$V_1\left(\frac{1}{6}\right) + 1 + V_2\left(\frac{1}{6}\right) - I = 0$$

$$\gg \frac{V_1}{6} + 1 + \frac{V}{6} - I = 0 \qquad \ldots\ldots\ldots\ldots \text{(i)}$$

Applying KVL in supernode,

$$V_1 - V = -12$$

$$\gg V_1 = -12 + V \qquad \ldots\ldots\ldots\ldots \text{(ii)}$$

(ii) in (i),

$$\frac{-12 + V}{6} + 1 + \frac{V}{6} - I = 0$$

$$\gg -2 + \frac{V}{6} + 1 + \frac{V}{6} - I = 0$$

$$\boxed{\gg I = \frac{V}{3} - 1}$$

To draw the graph, let's find two points (because it is a linear circuit and if we can find 2 points, we can draw a straight line through them to draw the IV characteristics)

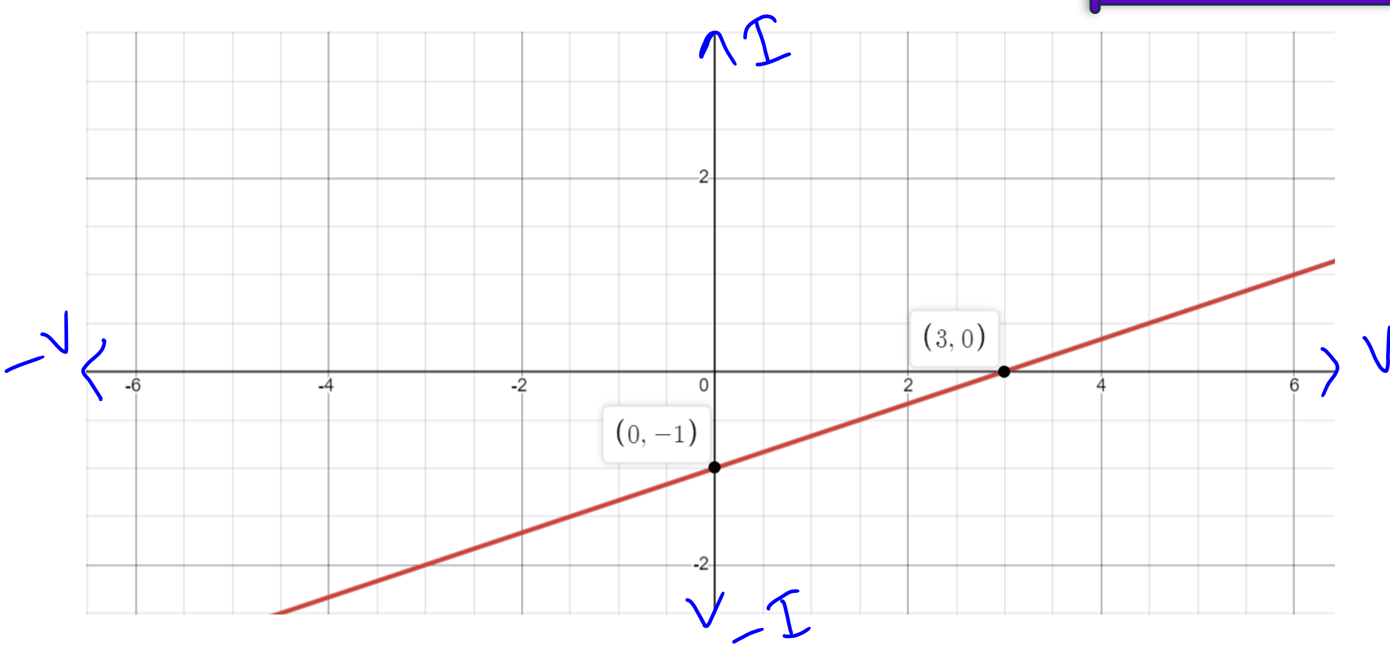
From the equation,

$$if\ I = 0, V = 3$$

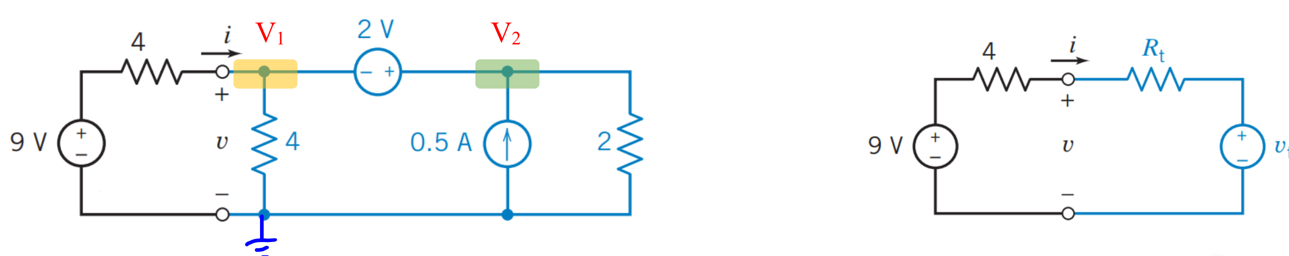
$$If\ V = 0, I = -1$$

So, we get two points, $(V_1, I_1) \equiv (3, 0)$ and $(V_2, I_2) \equiv (0, -1)$.

Do not forget to label the axis



Solution to Problem 4



In this problem, we have 2 circuits. One in the left has values of all the resistors and sources and the right circuit is missing value for R_t and v_t. As it is said that both the circuits are equivalent, the IV characteristics will be same (for blue marked portion for both circuits). So, we will find the IV characteristics (equation) from the left circuit and using that, will find the unknown parameters in right circuit.

From the left circuit,

$$V_1 = v$$

We can see the that there is a supernode between node 1 and 2. Applying KCL at supernode,

$$\frac{V_1}{4} - i + \frac{V_2}{2} - 0.5 = 0$$

$$\gg \frac{v}{4} - i + \frac{V_2}{2} - 0.5 = 0 \quad \text{...........} \text{ (i)}$$

Applying KVL at supernode,

$$V_1 - V_2 = -2$$

$$\gg v - V_2 = -2$$

$$\gg V_2 = v + 2 \qquad \text{...........} \text{ (ii)}$$

(ii) in (i),

$$\gg \frac{v}{4} - i + \frac{v+2}{2} - 0.5 = 0$$

$$\gg \frac{v}{4} - i + \frac{v}{2} + 1 - 0.5 = 0$$

$$i = \frac{3}{4}v + \frac{1}{2}$$

This IV charecteristic equation is in KCL form. To find out the unknown parameters in the right circuit, we need to make this equation into KVL form. So,

$$v - \frac{4}{3}i + \frac{2}{3} = 0 \quad \text{...........} \text{ (iii)}$$

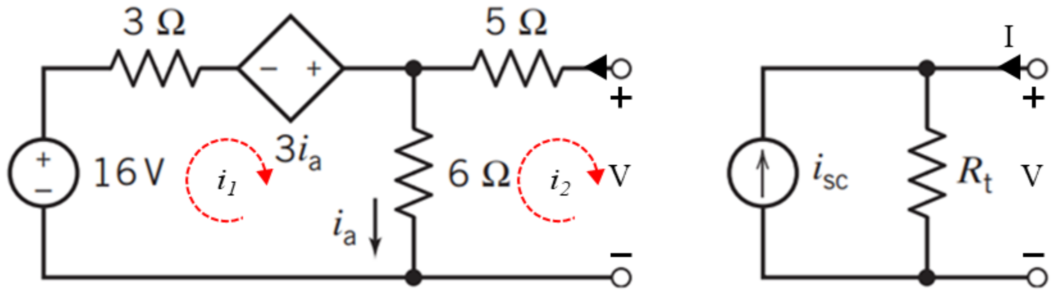
Applying KVL in the right circuit,

$$v - R_t i - v_t = 0 \quad \text{...........} \text{ (iv)}$$

Comparing (iii) and (iv),

$$\boxed{\begin{aligned} R_t &= \frac{4}{3}\ \Omega \\ v_t &= -\frac{2}{3}\,\text{V} \end{aligned}}$$

Solution to Problem 5



In this problem, we have 2 circuits. One in the left has values of all the resistors and sources and the right circuit is missing value for R_t and i_{sc}. As it is said that both the circuits are equivalent, the IV characteristics will be same (for blue marked portion for both circuits). So, we will find the IV characteristics (equation) from the left circuit and using that, will find the unknown parameters in right circuit.

From the left circuit,

$$i_a = i_1 - i_2$$

$$i_2 = -I$$

Applying KVL in mesh 1,

$$-16 + 3i_1 - 3i_a + 6(i_1 - i_2) = 0$$

$$\gg -16 + 3i_1 - 3(i_1 - i_2) + 6(i_1 - i_2) = 0$$

$$\gg -16 + 3i_1 + 3i_1 - 3i_2 = 0$$

$$\gg 6i_1 - 3i_2 = 16$$

$$\gg i_1 = \frac{1}{2}i_2 + \frac{16}{6}$$

Applying KVL in mesh 2,

$$6(i_2 - i_1) + 5i_2 + V = 0$$

$$\gg 6\left(i_2 - \frac{1}{2}i_2 - \frac{16}{6}\right) + 5i_2 + V = 0$$

$$\gg 3i_2 - 16 + 5i_2 + V = 0$$

$$\gg 8i_2 = -V + 16$$

$$\gg i_2 = -\frac{V}{8} + 2$$

$$\boxed{\gg I = \frac{V}{8} - 2} \quad \text{............ (i)}$$

This IV charecteristic equation is in KCL form. So applying KCL in the right circuit,

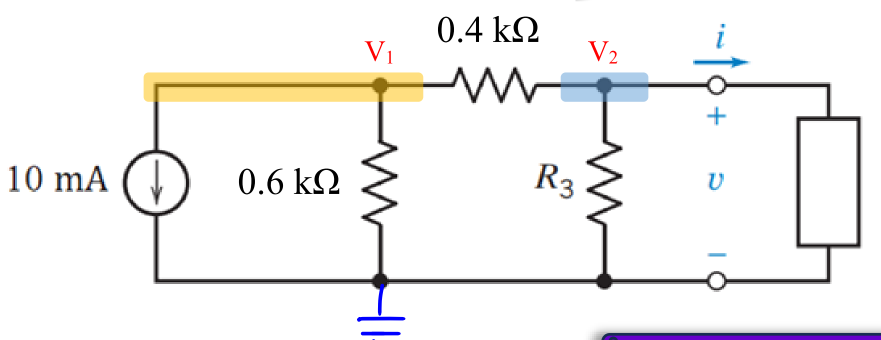
$$I = \frac{V}{R_t} - i_{sc} \quad \text{........... (ii)}$$

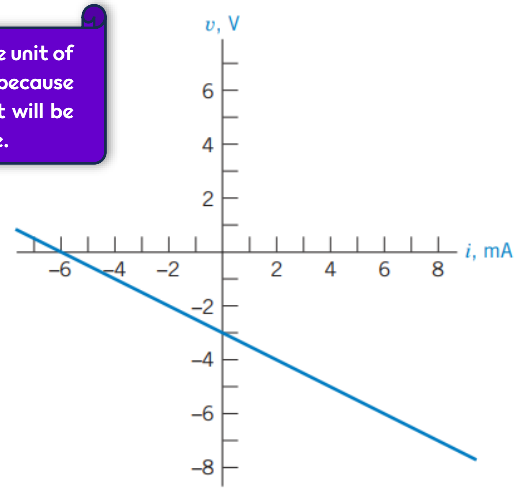
Comparing (i) and (ii),

$$\boxed{\begin{aligned} R_t &= 8\ \Omega \\ i_{sc} &= 2\ A \end{aligned}}$$

Solution to Problem 6

If you look closely, I converted the unit of the resistances into kΩ. I did it because the current source is in mA. So it will be easier to calculate using kΩ range.





Do not worry about current being in the x axis and voltage being in the y axis. I know it is unconventional. But if you keep the calculation according to the graph, it will be okay !

From the graph, we can find the IV characteristics (equation) and using that equation we can find the value of R_3

The equation of the IV characteristics can be found using the 2 points where the line intersects the x and y axis. Those points are:

$$(i_1, v_1) \equiv (-6, 0) \text{ and } (i_2, v_2) \equiv (0, -3).$$

The equation of the line:

$$\frac{v - v_1}{i - i_1} = \frac{v_1 - v_2}{i_1 - i_2}$$

$$\gg \frac{v - 0}{i - (-6)} = \frac{0 - (-3)}{-6 - 0}$$

$$\boxed{\gg i = -2v - 6} \quad \text{........... (i)}$$

Now, let us apply Nodal analysis in the circuit. We are hoping to find an equation at the last which has i, v and R_3. Then we will compare the equation to the IV equation and find out the value of R_3

From the circuit,

$$V_2 = v$$

Applying KCL at node 1,

$$V_1\left(\frac{1}{0.6} + \frac{1}{0.4}\right) - \frac{V_2}{0.4} + 10 = 0$$

$$\gg V_1\left(\frac{1}{0.6} + \frac{1}{0.4}\right) - \frac{v}{0.4} + 10 = 0$$

$$\gg V_1 = 0.6v - 2.4 \text{ .... (ii)}$$

Applying KCL at node 2,

$$V_2\left(\frac{1}{0.4} + \frac{1}{R_3}\right) - \frac{V_1}{0.4} + i = 0 \text{ .... (iii)}$$

(ii) in (iii),

$$v\left(\frac{1}{0.4} + \frac{1}{R_3}\right) - \frac{0.6v - 2.4}{0.4} + i = 0$$

$$\gg \frac{v}{0.4} + \frac{v}{R_3} - 1.5v + 60 + i = 0$$

$$\gg 2.5v + \frac{v}{R_3} - 1.5v + 6 + i = 0$$

$$\gg v + \frac{v}{R_3} + 6 + i = 0$$

$$\gg v\left(1 + \frac{1}{R_3}\right) + i + 6 = 0$$

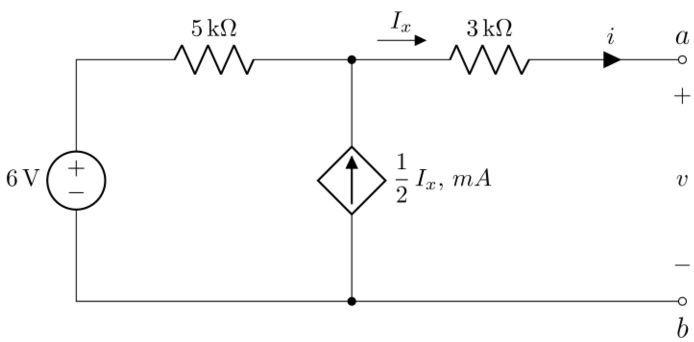
$$\gg i = -v\left(1 + \frac{1}{R_3}\right) - 6 \text{ .... (iv)}$$

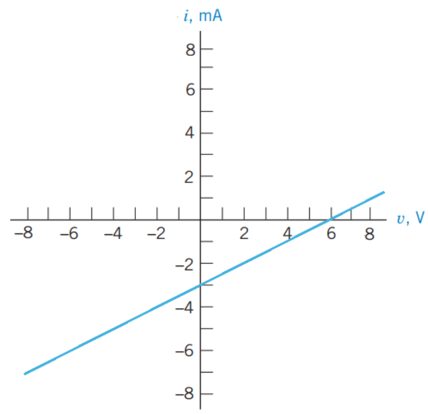
Comparing (i) and (iv),

$$1 + \frac{1}{R_3} = 2$$

$$\boxed{\gg R_3 = 1\ k\Omega}$$

Solution to Problem 7





From the graph, we can find the IV characteristics (equation) and using that equation we can find the value of the resistance of the dependent source.

The equation of the IV characteristics can be found using the 2 points where the line intersects the x and y axis. Those points are:

$$(v_1, i_1) \equiv (6, 0) \text{ and } (v_2, i_2) \equiv (0, -3).$$

The equation of the line:

$$\frac{v - v_1}{i - i_1} = \frac{v_1 - v_2}{i_1 - i_2}$$

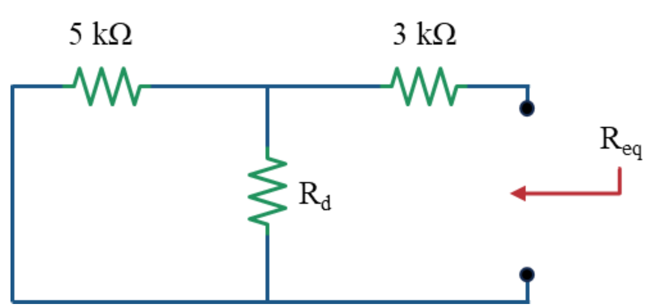
$$\gg \frac{v - 6}{i - 0} = \frac{6 - 0}{0 - (-3)}$$

$$\boxed{\gg i = \frac{1}{2}v - 3} \quad \ldots\ldots\ldots\ldots \text{(i)}$$

We can find the R_{eq} of the circuit from this equation,

$$R_{eq} = 2\ k\Omega \quad \ldots\ldots\ldots\ldots \text{(ii)}$$

To find the resistance contributed by the ideal source, let us find the R_{eq} from the circuit by replacing the dependent source with its resistance (let's say R_d) and replacing the voltage source with its ideal resistance which is zero. The circuit becomes as follows,



So,

$$R_{eq} = (5 \,||\, R_d) + 3$$

$$\gg R_{eq} = \frac{5 \times R_d}{5 + R_d} + 3 \quad \ldots\ldots\ldots\ldots \text{(iii)}$$

Comparing (ii) and (iii),

$$2 = \frac{5 \times R_d}{5 + R_d} + 3$$

$$\gg \frac{5 \times R_d}{5 + R_d} = -1$$

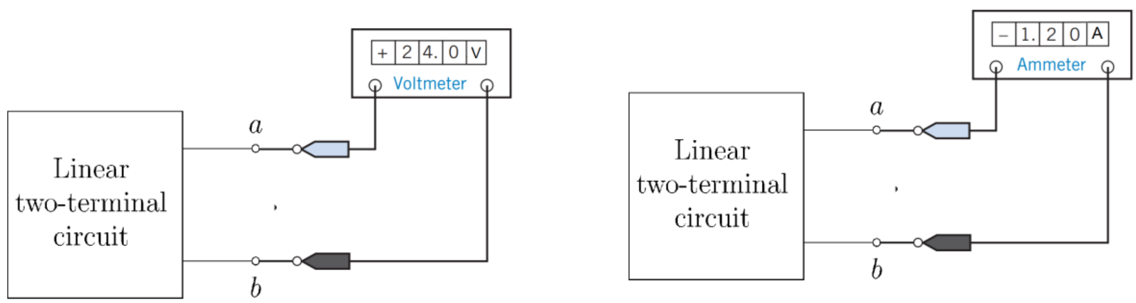
$$\gg 5R_d = -5 - R_d$$

$$\gg 6R_d = -5$$

$$\boxed{R_d = 0.83\ k\Omega}$$

Ignoring the minus sign

Solution to Problem 8



From voltmeter,

$$V = 24\ V$$

From Ammeter,

$$I = 1.2\ A$$

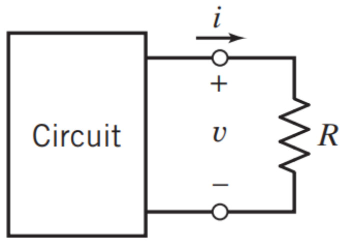
Using Ohm's law,

$$R = \frac{V}{I}$$

$$\gg R = \frac{24}{1.2}$$

$$\boxed{\gg R = 20\ \Omega}$$

Solution to Problem 9

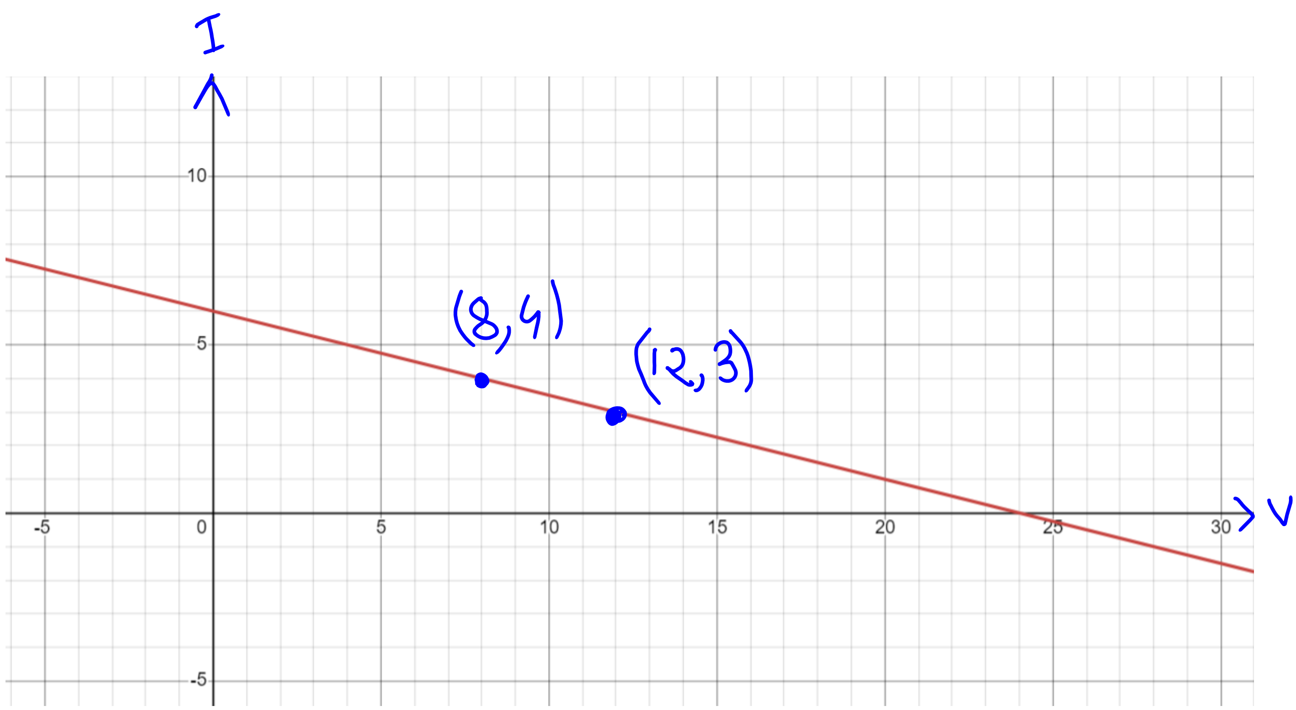


R	i
2 kΩ	4 mA
4 kΩ	3 mA

Let us complete the table first by findout out the value of voltages using Ohm's law, V = iR

R	i	V
2 kΩ	4 mA	8 V
4 kΩ	3 mA	12 V

i) From the table, we get two points, $(v_1, i_1) \equiv (8, 4)$ and $(v_2, i_2) \equiv (12, 3)$. So if we plot them,



The equation of the IV characteristics can be found using the 2 points.

The equation of the line:

$$\frac{v - v_1}{i - i_1} = \frac{v_1 - v_2}{i_1 - i_2}$$

$$\gg \frac{v - 8}{i - 4} = \frac{8 - 12}{4 - 3}$$

$$\boxed{\gg i = -\frac{1}{4}v + 6} \quad \text{..........} \ (1)$$

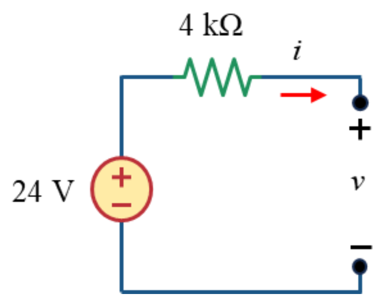
ii) We can see from the equation (1) that it is a KCL equation (Current i has no constant multiplied with it and voltage has). To make a circuit, we can use this KCL equation we can make it as KVL equation as well. (Which might be easier to draw, I prefer KVL equation to draw circuits)

So, from (1),

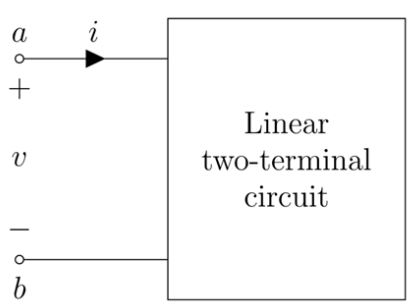
$$v + 4i - 24 = 0$$

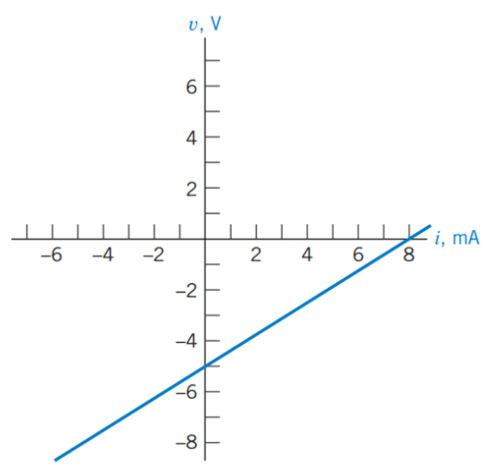
As this is a KVL equation, the 4 multiplied with current i, is the value of resistance in kΩ. And the constant 24 represents a voltage source.

So the circuit will be,



Solution to Problem 10





The equation of the IV characteristics can be found using the 2 points where the line intersects the x and y axis. Those points are:

$(i_1, v_1) \equiv (8, 0)$ and $(i_2, v_2) \equiv (0, -5)$.

The equation of the line:

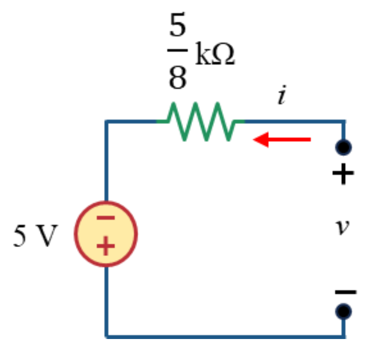
$$\frac{v - v_1}{i - i_1} = \frac{v_1 - v_2}{i_1 - i_2}$$

$$\gg \frac{v - 0}{i - 8} = \frac{0 - (-5)}{8 - 0}$$

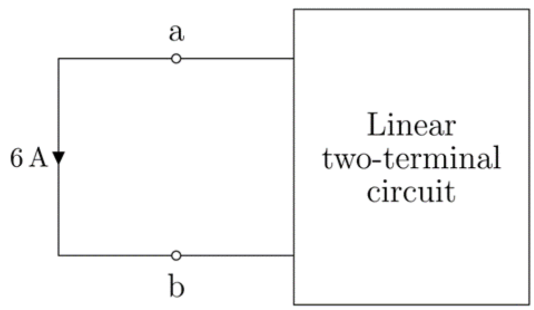
$$\boxed{\gg v - \frac{5}{8}i + 5 = 0} \quad \ldots\ldots\ldots\ldots \text{(i)}$$

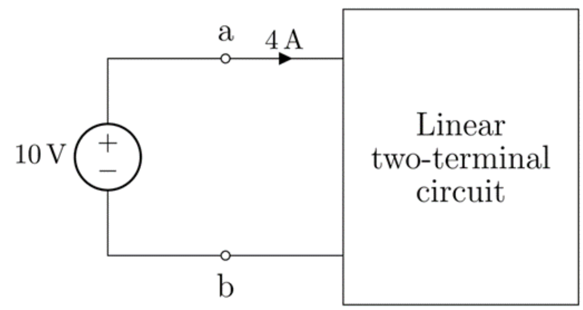
So, the circuit will be,

As this is a KVL equation, the 5/8 multiplied with current i, is the value of resistance in kΩ. And the constant 5 represents a voltage source.



Solution to Problem 11





From the circuits, we get two data points for I-V characteristics,

$(v_1, i_1) \equiv (0, -6)$ and $(v_2, i_2) \equiv (10, 4)$.

The equation of the I-V characteristics,

$$\frac{v - v_1}{i - i_1} = \frac{v_1 - v_2}{i_1 - i_2}$$

$$\gg \frac{v - 0}{i - (-6)} = \frac{0 - 10}{-6 - 4}$$

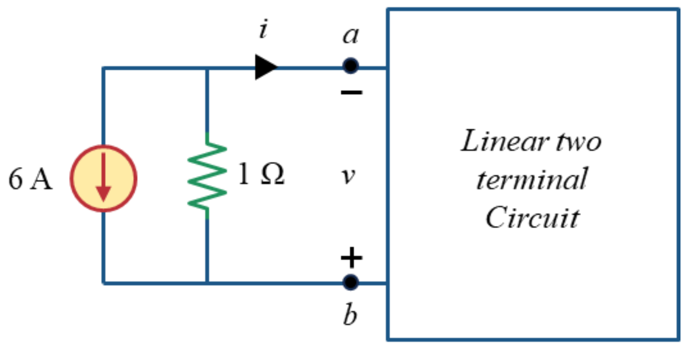
$$\boxed{\gg i = v - 6}$$

As the equation does not have any constant multiplied with both i and v, we can consider the equation as KCL or KVL as our wish. We can construct two circuits from two points of views (KCL and KVL equation).

Considering the equation as KCL,

$$i + 6 = v$$

Constructing the circuit as per the equation,



Considering the equation as KVL,

$$v - i - 6 = 0$$

Constructing the circuit as per the equation,